

LAUD ASANTE

Data Scientist | Machine Learning · Forecasting · Optimisation · MLOps

Kingston upon Hull, UK | Open to relocation and hybrid across the UK

07377 261 459 | asantelaud@gmail.com

linkedin.com/in/laud-asante-938382103 | github.com/Laud1ens

Portfolio: laud-asante-web-production.up.railway.app

Right to work in the UK | No sponsorship required | Available immediately

PROFESSIONAL SUMMARY

MSc Artificial Intelligence and Data Science candidate (University of Hull, completing Sept 2026) with six years of commercial and operational management experience and a quantitative foundation in Mathematics and Economics. Builds end-to-end ML pipelines, forecasting models and optimisation-driven classifiers in Python and SQL, and delivers Power BI dashboards for non-technical stakeholders. Translates technical findings into plain-language explanations for non-technical audiences. Maintains an open, documented portfolio of ML projects on GitHub.

CORE TECHNICAL SKILLS

Languages: Python (Advanced), SQL / MySQL, Bash

ML / DL: Scikit-learn, TensorFlow/Keras, XGBoost, LightGBM, PyTorch, GNN/GAT, LSTM/GRU, Transformers

Forecasting: SARIMA, SARIMAX, Time-Series Analysis, Demand Planning

NLP: NLTK, spaCy, TF-IDF, Transformer fine-tuning (MiniLM), Sentiment Analysis

Statistics: Hypothesis Testing, A/B Testing, Bayesian Inference, McNemar's Test, Regression

Data Eng. & BI: Pandas, NumPy, SQLAlchemy, ETL Pipelines, Power BI, Matplotlib/Seaborn

Optimisation: Hyperparameter Tuning (Grid/Random/Bayesian), Feature Selection, Cost-sensitive Thresholding

MLOps / Dev: Git/GitHub, Jupyter, Docker (familiar), MLflow (familiar), CI/CD awareness, AWS (awareness)

SELECTED PROJECTS

MSc Dissertation (in progress): US County-Level COVID-19 Forecasting

SARIMA · SARIMAX · XGBoost · LSTM · GAT-LSTM · Supervised by Dr. Tongxin Chen · Submitting Aug 2026

- Building a four-notebook, seven-model forecasting pipeline (SARIMA, SARIMAX, XGBoost, LSTM, GAT-LSTM variants) benchmarked across 3,219 US counties.
- Current headline result: SARIMA leads with $R^2 = 0.745$, outperforming the more complex GAT-LSTM architectures, an evidence-based case for choosing the simpler model that actually generalises.

Manufacturing Predictive Maintenance (AI4I 2020)

Python · Scikit-learn · Gradient Boosting · Imbalanced Classification

- Built a machine-failure risk classifier on 10,000 milling-machine sensor records with a 3.39% failure rate, comparing Random Forest, Gradient Boosting and Logistic Regression on precision, recall, F1, ROC-AUC and average precision.
- Outcome: Gradient Boosting selected on F1 at 94.7% precision and 79.4% recall, with cross-validated recall reported as a range (0.761 ± 0.057 over 5 stratified folds) rather than a single figure, and a decision-threshold sweep in place of an inherited default cutoff.
- Cross-checked feature importance two ways and reported the disagreement rather than a single headline driver; the actionable finding was that failure rate roughly triples (2.2% to 6.1%) once tool wear passes its critical band. Recall held at 76-84% across all machine grades, including the low-grade units that fail at nearly twice the rate.

NexaHeat FX: Forex Intelligence Platform

Python · Twelve Data API · Claude/Groq/Gemini · Solo build, ongoing

- Designed and built a browser-based FX intelligence tool from scratch: live market data integrated with three LLM providers for real-time commentary and signals, plus a currency-strength heatmap and order-flow engine.
- Outcome: Shipped a working, end-to-end multi-API product independently; now scoping a market-prediction layer (XGBoost, progressing toward a Temporal Fusion Transformer).

Sarcasm Detection: Deployment-Ready Model Comparison

Python · TensorFlow/Keras · Scikit-learn · Transformers (MiniLM)

- Compared six approaches (Naive Bayes, Logistic Regression, LSTM, GRU, MiniLM frozen/fine-tuned) for sarcasm detection on 28,503 headlines, then stress-tested every model on 24 out-of-domain sentences.
- Outcome: Fine-tuned MiniLM reached 92% in-domain accuracy (0.974 AUC-ROC) but collapsed to 50% out-of-domain, identical to the weaker LSTM; significance confirmed via McNemar's test ($p < 0.001$).

EDUCATION

MSc Artificial Intelligence and Data Science | University of Hull *Sept 2025 - Sept 2026*

- Modules: Machine Learning for Healthcare, Big Data Analytics, Neural Network Design, Advanced Statistical Modelling, Bioinformatics, Scalable AI Systems.

Data Analysis Bootcamp | Generation Ghana *Aug - Oct 2024*

- One-month intensive with inbuilt data analyst mentoring; Excel, SQL, Python, Power BI; hands-on EDA and dashboard delivery.

BA Mathematics and Economics | University of Ghana *Completed May 2015*

- Quantitative analysis, econometrics, probability theory, linear algebra.

PROFESSIONAL EXPERIENCE

Strategic Area Sales Manager | Sweet Nutrition Limited *Jun 2022 - Oct 2025*

- AI-assisted trend analysis and Excel forecasting across 5 territories and 200+ accounts; produced analytical reports and KPI dashboards for senior leadership across 3 product categories.

Branch Sales Manager | Classfam Pharmaceuticals *2021 - 2022*

- Managed branch operations in a regulated pharmaceutical environment; applied data-integrity and audit-trail standards directly analogous to production ML data governance.

Territory Sales Supervisor / Manager | Sunda Invest Limited *2018 - 2021*

- Supervised cross-functional field teams against KPI targets; analysed sales throughput to identify territory-level efficiency gaps.

CERTIFICATIONS & CPD

- IBM Data Science Professional Certificate (10-course) - IBM/Coursera, 2024
- Machine Learning Specialization (3-course) - Andrew Ng/DeepLearning.AI, 2024
- Google Data Analytics Professional Certificate - Google/Coursera, 2024
- Deep Learning Specialization (5-course) - DeepLearning.AI/Coursera, 2024-25
- Google Developer Program - 9 ML/AI knowledge badges, Google for Developers, 2025